

MD Analysis Pipeline

Quick Start Guide | analyze_md.py

Step 1. Set up your folder

Create a folder called `analysis/` anywhere on your computer.
Copy your GROMACS simulation files AND the script files into it:

```
analysis/          <- CREATE this folder
|
|-- md_0.1.tpr      <- copy your GROMACS files here
|-- md_0.1.xtc
|-- md_0.1.edr
|-- md_0.1.gro
|-- index.ndx
|
|-- install.py      <- copy the script files here too
|-- analyze_md.py
|-- md_config.ini
|
|-- results/        <- CREATED automatically
|   |-- summary_panel.png
|   |-- rmsd.png    ... (all figures)
|   |-- tables/
|       |-- rmsd.csv (all data tables)
|       |-- analysis_report.txt
```

Step 2. Install (run once, takes 1-2 minutes)

Open a terminal, go to the analysis folder, and type:

```
python install.py
```

The script downloads and installs all required packages automatically. You only need to do this once.

Step 3. Edit md_config.ini (2 settings only)

Open `md_config.ini` in Notepad (Windows) or TextEdit (Mac).

Change only these 2 values at the top of the file:

Setting	Section	What to put	Example
lig_resname	[system]	Ligand 3-letter residue name	LIG
gatekeeper_res	[system]	Key residue number to track	315

Not sure of your ligand name? Open the `.gro` file in Notepad/TextEdit and look for a name that is NOT: WAT SOL HOH NA CL MG K
That remaining name is your ligand.

Everything else in `md_config.ini` already has sensible defaults.
Only edit colours, DPI, or figure size if you want custom output.

Step 4. Run the analysis

```
python analyze_md.py
```

The script prints a progress log. Results are saved in two places:

Figures -> analysis/results/

Tables -> analysis/results/tables/

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Output Files

Figures (PNG) saved in results/

File	What it shows
energy_potential.png	Potential energy -- should be stable
energy_kinetic.png	Kinetic energy over time
energy_total_energy.png	Total energy over time
energy_temperature.png	Temperature -- should stay near 300 K
energy_pressure.png	Pressure over time
energy_volume.png	Box volume over time
energy_density.png	System density -- ~1000 kg/m3 for water
rmsd.png	Backbone & ligand RMSD -- stable = good run
rmsf.png	Per-residue flexibility -- peaks = flexible loops
gyration.png	Protein compactness over time
hbonds.png	Protein-ligand H-bonds per frame
gatekeeper_distance.png	Ligand to gatekeeper residue distance
pocket_contacts.png	Binding-pocket residues by contact frequency
summary_panel.png	4-panel overview -- open this FIRST

Tables (CSV) and text report saved in results/tables/

File	Contents
rmsd.csv	Time, backbone RMSD, ligand RMSD
rmsf.csv	Residue number, name, flexibility (RMSF)
hbonds_count.csv	H-bond count per simulation frame
hbonds_pairs.csv	Each H-bond pair with occupancy %
gatekeeper_distance.csv	Min distance to gatekeeper per frame
pocket_contacts.csv	Each contact residue with occupancy %
analysis_report.txt	Full text summary with all key numbers

Common Problems

Problem	Solution
Required file not found	GROMACS files must be in the SAME folder as analyze_md.py
gmx: command not found	Load GROMACS: module load gromacs (HPC) or install it
Ligand RMSD is very large	Check lig_resname in md_config.ini -- must match .gro exactly
Wrong index group detected	Add [groups] override to bottom of md_config.ini
Figures too small or text too small	Increase width, height, or font_scale in md_config.ini

Optional Command-Line Flags

Flag	Example	What it does
--dpi	--dpi 300	High-res figures for publication
--width/--height	--width 14 --height 5	Change figure size
--ligand-color	--ligand-color "red"	Change ligand colour
--out-dir	--out-dir /data/results	Save results elsewhere
--sim-dir	--sim-dir /data/run2	Files in a different folder
--config	--config /path/config.ini	Use a different config file

All options: `python analyze_md.py --help` | Full docs: `USER_MANUAL.md`